MIL-P-7968A
23 March 1965
SUPERSEDING
MIL-P-7968
29 August 1952

MILITARY SPECIFICATION

POSITION INDICATING ASSEMBLIES; ELECTRICAL, WHEELS AND FLAPS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers design and all performance requirements for procurement of wheel and flap position indicating assemblies.

1.2 Classification - The position indicating assemblies shall be of the following types:

	Transmitter	Indicator
Type I	AN5785-1	MS90367-1
Type II	AN5785-1	MS90367-2

2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on the date of the invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

DD-G-451	Glass, Flat and Corrugated, for Glazing, Mirrors, and Other Uses
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Plating, Electrodeposited, Requirements for

THIS DOCUMENT CONTAINS 22 PAGES.

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Military

MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of
MIL-N-3336	Nut, Self Locking, Instrument Mounting
MIL-C-5015	Connectors, Electric, "AN" Type
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-R-22973	Reliability Index Determination for Avionic Equipment Models, General Specification for
MIL-L-25142	Luminescent Material, Fluorescent
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

Fed. Std. No. 595	Colors

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Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS28105	Window, Dial - Aircraft Instrument Cover, Glass
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33586	Metals, Definition of Dissimilar
MS90367	Indicator-Electrical, Wheels and Flaps Position
AN5785	Transmitter-Electrical, Wheels and Flaps Position

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification - The position indicating assembly furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Selection of Government documents - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143, except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts - Commercial parts having suitable properties shall be used where, on the date of invitations for bids, there are no suitable standard parts. In any case, commercial parts such as screws, bolts, nuts, cotter pins, having suitable properties, may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

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(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.2.2 Standard parts - With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Critical materials - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Nonmagnetic materials - Nonmagnetic materials shall be used for all parts of the position indicating assembly except where magnetic materials are essential.

3.3.3 Metals - Metals shall be of the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt spray, or atmospheric conditions to which the position indicating assembly may be subjected when in storage or during normal service life.

3.3.3.1 Dissimilar metals - Dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coatings.

3.3.3.2 Magnesium alloy parts - Magnesium alloy parts shall be treated in accordance with MIL-M-3171. When abrasion resistance is a factor, an anodic treatment approved by the procuring activity shall be used.

3.3.3.3 Aluminum alloy parts - Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Small holes, threads, and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.3.3.4 Iron and steel parts - Iron and steel parts which are in hermetically sealed cases shall be cadmium plated in accordance with QQ-P-416. Parts which are not in sealed cases shall be chromium, nickel, or zinc plated in accordance with QQ-C-320, QQ-N-290, and QQ-Z-325, respectively. Parts in a confined space in the presence of organic material shall be tin-cadmium plated.

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3.3.4 Protective treatment - When materials are used in the construction of the position indicating assemblies that are subject to atmospheric or environmental conditions likely to cause corrosion in normal service life, they shall be protected against corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip, or scale during normal service life or are affected by extremes of atmospheric or environmental conditions shall not be used.

3.3.5 Fungus-proof materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used, they shall be treated with a fungicidal agent acceptable to the procuring activity. If used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.6 Fumes and vapors - Materials used in the construction of the position indicating assemblies shall not produce corrosive, deleterious, or toxic fumes or vapors under the conditions specified herein.

3.4 Design and construction - Each position indicating assembly shall consist of one indicator and one position transmitter. The indicator shall conform to MS90367. The transmitter shall conform to AN5785. The design shall provide access to the indicator mechanism for adjustment, inspection, and repair without removing the cover glass or bezel from the case. The transmitter shall be designed so that moisture cannot accumulate around the operating shaft. The indicator and transmitter shall be so constructed that no parts will work loose in service and will withstand the normal shocks, vibrations, and such other conditions as are incident to service without failure, shipping, storage, and installation.

3.4.1 Transmitter stops - The transmitter shall be provided with stops which will permit a minimum overtravel of 45 degrees at each end of the operating resistance (180 degrees minimum shaft rotation). In addition, the shaft and case shall be marked so that the shaft can be correctly aligned with the operating resistance.

3.4.2 Maintenance - The design shall be reasonably simple to facilitate as much as possible disassembly, repair or overhaul, service maintenance, and reassembly using those tools and items of maintenance equipment which are normally available as commercial standards.

3.4.3 Case - The indicator case shall conform to MS90367. The transmitter case shall conform to AN5785. Each case shall be made of nonferrous, low density metal, uniform in texture and shall have a smooth surface. Each case shall be finished with a lusterless black material, Color No. 37038 of Fed. Std. No. 595. The finishing material shall be of a durable type to withstand usage encountered in service.

3.4.4 Weight - The indicator shall be so designed that when completely assembled, the weight shall not exceed 1.1 pounds. The transmitter shall be so designed that when completely assembled, the weight shall not exceed 0.4 pound.

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3.4.5 Screw threads - Screw threads 0.060 inch in diameter or larger, shall be in accordance with MIL-S-7742.

3.5 Electrical requirements -

3.5.1 Primary input power requirements - The position indicating assemblies shall perform as specified herein when connected to a 28-volt dc power source with characteristics as defined in MIL-STD-704, Category B.

3.5.2 Undervoltage protection - The position indicating assemblies shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage is within the specified limits.

3.5.3 Electric connectors - The electric connectors used shall conform to MIL-C-5015. The connector for the indicator shall conform to MS90367. The connector for the transmitter shall conform to AN5785. The connector code number markings as specified in MIL-E-5400 shall be legibly and permanently applied to the case or connector.

3.5.4 Wiring - The wiring to the electric connector of the indicator and transmitter shall be as shown in Figure 1. Internal wiring shall be color coded, neat, and accomplished in such a manner that individual wires may be easily traced. Wires shall be secured mechanically to the terminals prior to soldering. The internal wiring of the transmitter shall conform to Figure 1 and Table I. The wiring shall be such that clockwise rotation (as viewed from the shaft end) will move the brush from D to A. It shall be possible to use the transmitter for either clockwise or counterclockwise operation (as viewed from the shaft end). For flap "Down" indication with clockwise rotation of the transmitter shaft, the wiring connections shall be as shown in Figure 1. The wiring connections for counterclockwise transmitter shaft rotation (flaps "Down") shall be as shown in Figure 1 except that the transmitter leads shall be reversed. The transmitter resistance windings shall be coated with a suitable cement, except at the line of contact, to prevent displacement of the winding. The resistance winding at the point of contact shall be smooth and free of surface irregularities.

TABLE I

Transmitter Windings

Winding	Resistance Ohms	Angular Position of Coil Taps-Degrees
A-D	196 ±15 percent	90 ±2
A-B	52 ±15 percent	17 +2 -1

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TABLE I (Cont'd)

Winding	Resistance Ohms	Angular Position of Coil Taps-Degrees
B-C	92 ±15 percent	56 ±3
C-D	52 ±15 percent	17 +2 -1
R_1	275 ±10 percent	----

3.6 Dial - The dial cut-outs and outline of wheels and flaps on tabs shall conform to Figure 2 and Figure 3. The dial shall be securely mounted in such a manner that it will not loosen or slip when the indicator is vibrated. If screws are used, they shall be so located that they will not interfere with the proper location of any dial marking.

3.6.1 Visibility of dial - The tab, all graduations, numerals, and other specified markings on the dial shall be visible from any point within a frustum of a cone whose side makes an angle of 30 degrees with a perpendicular to the dial and whose small diameter is the aperture of the case.

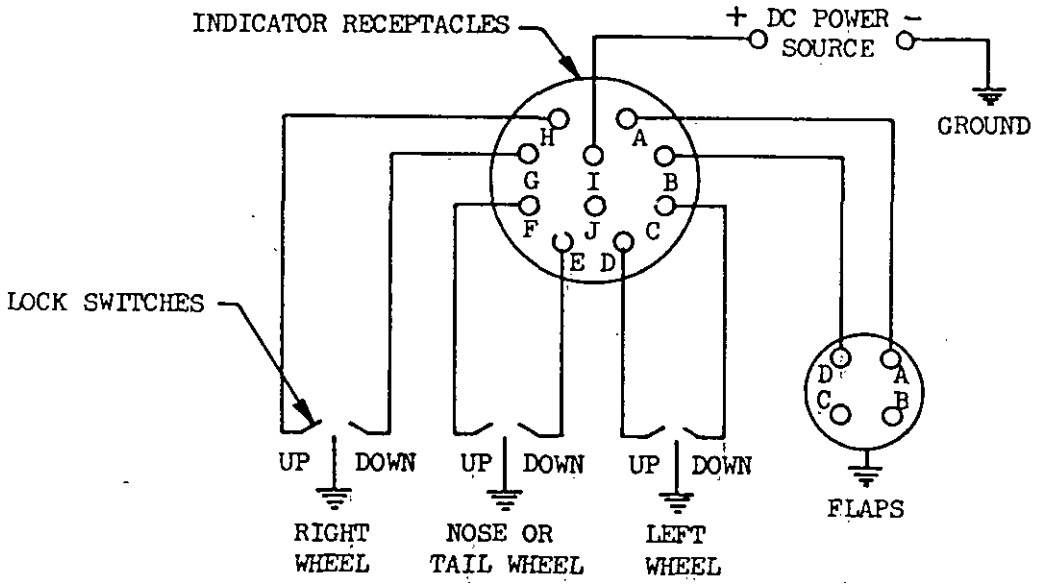
3.6.2 Dial marking - The indicator dial shall be marked as shown in Figure 3. Unless otherwise specified, all visible portions of the dial shall be lusterless black Color No. 37038 of Fed. Std. No. 595 and all markings on the dial shall be lusterless white Color No. 37875 of Fed. Std. No. 595 or fluorescent luminescent material conforming to MIL-L-25142. All markings shall be durable to withstand usage encountered in service. There shall be no change in the color of the markings as a result of subjecting the indicator to the tests specified herein. The form of the numerals and letters shall be in accordance with MS33558. Numerals shall distinctly indicate the graduation to which each applies. When several letters are used in one group, the space between the letters shall be not less than 1/64 inch.

3.6.3 Pointer and movable tabs - The pointer and movable tabs shall be as light as practicable and sufficiently rigid to prevent oscillation under vibration.

3.6.4 "OFF" alarm - The indicator shall have an "OFF" alarm sign on the flap position indicator which will appear in the flap cut-out or in an adjoining cut-out when no power is being supplied to the indicator.

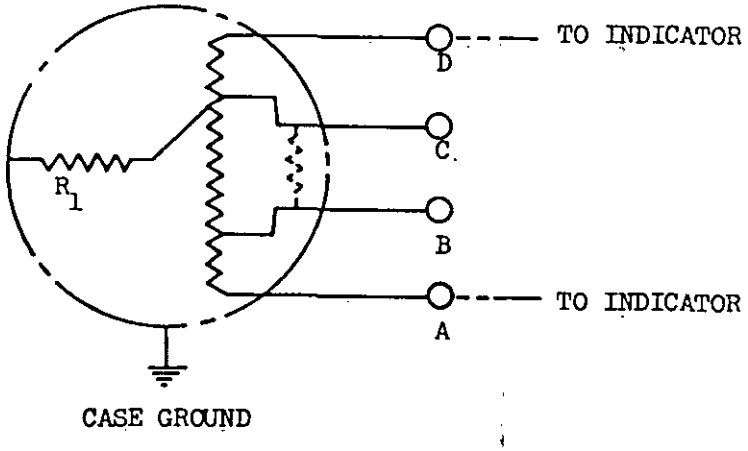
3.7 Cover glass - The cover glass for the indicator shall be in accordance with DD-G-451, Type II, Quality AA. Any flaws permitted by DD-G-451 shall not interfere with reading of the dial. The dimensions of the cover glass shall be in accordance with MS28105.

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CONNECTION SHOWN IS FOR FLAP "DOWN" INDICATION WITH CLOCKWISE ROTATION OF TRANSMITTER SHAFT, AS VIEWED FROM SHAFT END. GROUND TRANSMITTER HOUSING FOR NEGATIVE BATTERY RETURN CIRCUIT.

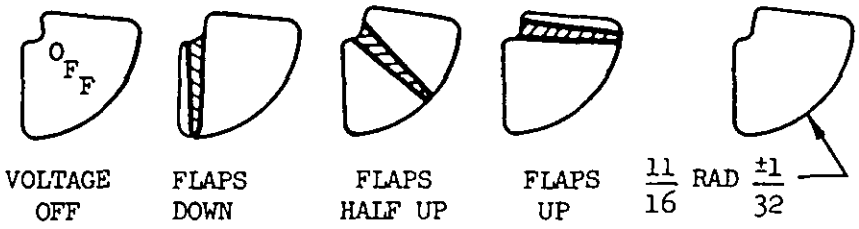
POSITION INDICATING ASSEMBLY EXTERNAL WIRING DIAGRAM



POSITION TRANSMITTER INTERNAL WIRING DIAGRAM

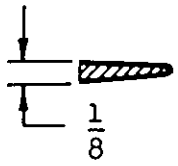
FIGURE 1. Electrical Connections

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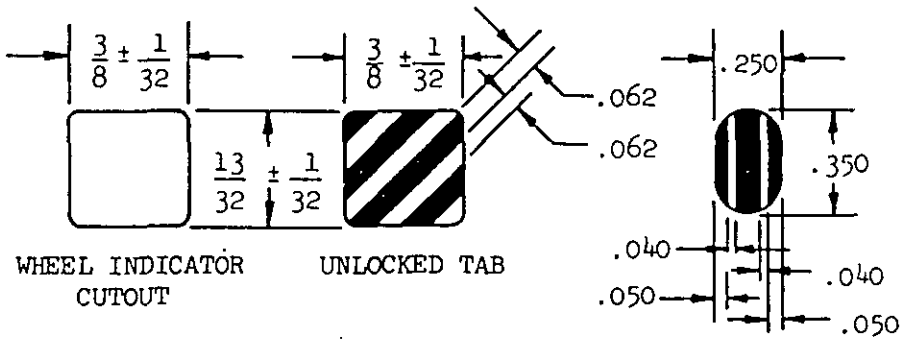


FLAP INDICATOR POSITIONS

OPENING FOR FLAP POSITION INDICATOR



INDICATOR TAB FOR FLAPS



WHEEL INDICATOR CUTOUT

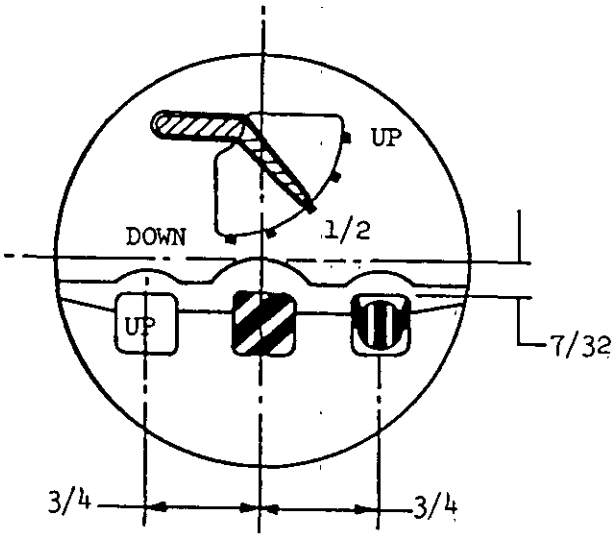
UNLOCKED TAB

WHEEL DIMENSIONS

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED,
TOLERANCES: FRACTIONS ±1/64, DECIMALS ±0.010.

FIGURE 2. Dimensions of Tabs and Cutouts

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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: FRACTIONS ±1/32, DECIMALS ±0.010

MARKING	HEIGHT OR LENGTH	WIDTH OF LINE OR GRADUATION ±.005
THE FOLLOWING SHALL BE FINISHED IN FLUORESCENT LUMINESCENT OR LUSTERLESS WHITE MATERIAL.		
PROFILE OF WING AND FLAP	--	1/32
UNSHADED PART OF WHEEL UNLOCKED TAB	--	SEE FIGURE 2
SHADED PART OF WHEEL	--	SEE FIGURE 2
WORD "OFF" (FLAPS TAB)	1/16	1/64
WORD "UP" (WHEEL TAB)	1/8	1/32
WORDS "UP" AND "DOWN"	1/16	1/64
NUMERAL "1/2"	1/16	--
FLAP GRADUATIONS	1/16	1/32
WHEEL SUPPORTS	--	1/32
CROSS HATCH LINES OF WING AND FLAP	--	--
LINES OF PLANE PROFILE	--	1/32
THE FOLLOWING SHALL BE FINISHED IN LUSTERLESS BLACK MATERIAL		
SHADED PART OF WHEEL UNLOCKED TAB		SEE FIGURE 2

FIGURE 3. Dial

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3.7.1 Cover glass mounting - The distance from the inner surface of the cover glass to the surface of the dial on which the marking is applied shall not exceed 0.0625 inch.

3.8 Performance - The position indicating assemblies shall perform satisfactorily when subjected to the tests specified in Section 4 of this specification.

3.8.1 Special reliability design - The design of the position indicating assemblies shall be planned, analyzed and the reliability reported (see 6.3) as outlined in MIL-R-22973. The reliability goal, expressed in mean-time-between-failure (MTBF), shall be 1000 hours. The position indicating assemblies shall be tested in accordance with Test Level III of MIL-R-22973.

3.9 Identification of product -

3.9.1 Nameplate - A nameplate shall be securely attached to the exterior of the case of each indicator and transmitter and shall be marked in accordance with the requirements of MIL-STD-130 except that the FSN marking shall be omitted.

3.9.2 Manufacturer's part number - The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number, including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.10 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of MIL-D-70327.

3.11 Mounting hardware - The contractor shall furnish mounting screws and removable self-locking nuts conforming to MIL-N-3336 in sufficient quantity to install each indicator. The screws shall be round head brass machine screws having a durable, lusterless black, oxidized or nickel finish. The length of the screws shall be sufficient to mount the indicator on panels up to 3/8 inch thick.

3.11.1 Envelope - An envelope furnished by the contractor shall be packaged with each indicator and shall be marked as follows:

IMPORTANT
THIS ENVELOPE CONTAINS MOUNTING SCREWS AND NUTS

3.12 Workmanship - The position indicating assembly, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding and brazing, and freedom of parts from burrs and sharp edges.

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3.12.1 Dimensions - Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. When dimensions and tolerances affect the interchageability, operation, or performance of the position indicating assembly, they shall be held or limited accordingly.

3.12.2 Fabrication - Machining, drilling, and forming shall be done with the use of accurate templates, jigs, or gages.

3.12.3 Screw assemblies - Assembly screws shall be tight. The word "tight" means that the screws cannot be appreciably tightened without damage or injury to the screw threads or heads.

3.12.4 Riveting - Riveting operations shall be carefully performed in order to insure that the rivets are tight. Rivet heads shall be full size and concentric with the rivet body. Dimensions of the rivets shall be such that the holes will be filled. Loose, cracked, or badly formed rivets, including protruding countersunk rivet heads, shall be replaced before acceptance of the position indicating assembly. Excess metal shall be removed from countersunk rivet heads as a customary operation. This does not cover eyelets or hollow tubular rivets.

3.12.5 Gears - Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be as free as possible from backlash.

3.12.6 Cleanup - The position indicating assemblies shall be thoroughly cleaned of loose, spattered, or excess solder, resin flash which may crumble, and other foreign material after final assembly.

3.12.7 Wire stripping - Wire stripping shall be done without nicking or otherwise damaging the wire.

3.12.8 Soldering - Soldering shall be performed in accordance with MIL-S-6872.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of tests - The inspection and testing of the position indicating assemblies shall be classified as follows:

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(a) Qualification tests: Qualification tests are those tests performed on sample position indicating assemblies submitted for qualification as a satisfactory product.

(b) Acceptance tests: Acceptance tests are those tests performed on position indicating assemblies manufactured and submitted for acceptance under contract.

4.3 Qualification tests - The qualification tests of the position indicating assemblies shall consist of all of the tests of this specification specified under the paragraph headed Test methods.

4.3.1 Qualification test sample instructions - Qualification test samples shall consist of three position indicating assemblies manufactured in accordance with this specification. The position indicating assemblies submitted for qualification testing shall not have been previously tested except for the Individual tests. The samples shall be forwarded to the testing laboratory designated in the Letter of Authorization.

4.3.1.1 Test sample identification - The test samples shall be plainly identified by securely attached, durable tags marked with the following information:

Sample for Qualification Testing
POSITION INDICATING ASSEMBLIES; ELECTRICAL, WHEELS AND FLAPS
Submitted by (Manufacturer's name, date) for qualification testing in accordance with MIL-P-7968
under authorization (reference letter authorizing tests)
Type
Manufacturer's Part Number

4.4 Acceptance tests - The acceptance tests shall consist of the Individual tests, the Sampling plan tests, and the Reliability assurance tests of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests except that the Sampling plan B tests shall be conducted at a Government laboratory designated by the procuring activity. Acceptance tests, except for Sampling plan B tests, shall be under the supervision of the Government inspector. The contractor shall furnish test reports (see 6.3) showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

4.4.1 Individual tests - Each position indicating assembly submitted for acceptance shall be subjected to the Individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each position indicating assembly accepted shall have passed the following tests, conducted in the order listed:

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Examination of product
Power off
Insulation resistance
Operation
Friction
Reduced voltage operation

4.4.2 Sampling plan tests - The Sampling plan tests shall consist of Sampling plan A and Sampling plan B tests. The test samples selected for sampling tests shall first have passed the Individual tests. When a position indicating assembly fails to meet the requirements of this specification, assemblies still on hand or subsequently produced shall not be accepted until the extent and cause of failure have been determined and corrected. The test samples which have been subjected to Sampling plan A tests shall not be delivered on contract until they have been refurbished, resubmitted and passed all the Individual tests. Test samples which have been subjected to the Sampling plan B tests shall not be delivered on contract.

4.4.2.1 Sampling plan A sample selection - Sampling plan A samples shall be selected at random in accordance with the following schedule:

Quantity of Position Indicating Assemblies Offered for Acceptance		Quantity to be Selected for Testing
First	15	*
Next	50	1
Next	75	1
Next	100	1
Each additional or fraction thereof	200	1

*NOTE: When Sampling plan B is invoked, the quantity shall be zero.
When Sampling plan B is to be omitted, the quantity shall be one.

4.4.2.1.1 Sampling plan A tests - Each sample selected for Sampling plan A testing shall be subjected to the following tests, conducted in the order listed:

Magnetic effect
High temperature operation
Low temperature operation
Position error
Damping
Vibration (resonance)

4.4.2.2 Sampling plan B instructions - Two position indicating assemblies shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. These samples shall be forwarded to a Government laboratory designated by the procuring activity. Each sample shall be plainly identified by a securely attached, durable tag marked with the following information:

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POSITION INDICATING ASSEMBLIES; ELECTRICAL, WHEELS AND FLAPS
Submitted by (Manufacturer's name, date) for production acceptance Sampling plan B testing, in accordance with Contract or Order No. _____
Manufacturer's part number _____

4.4.2.2.1 Sampling plan B tests - Each sample selected for Sampling plan B testing shall be subjected to the following tests, conducted in the order listed:

Sampling plan A
Vibration (cycling)
Radio noise interference
Transmitter stop
High temperature exposure
Low temperature exposure
Transmitter cycling
Indicator cycling
Salt spray
Mounting lugs
Internal examination

4.4.3 Special reliability design test -

4.4.3.1 Mean-time-between-failures (MTBF) determination - Tests shall be performed to determine the MTBF of the position indicating assemblies. MIL-R-22973 shall be used as a guide for the determination. The procedure to be followed is outlined in the following paragraphs.

4.4.3.2 Test procedure outline - A detailed reliability test procedure outline shall be submitted to, and approved by, the procuring activity prior to the initiation of the tests. The test procedure must include but is not limited to:

(a) Duty cycle.
(b) Performance characteristics to be measured.
(c) Failure criteria.

4.4.3.3 Test procedure - Three position indicating assemblies shall be utilized for the reliability determination which shall be performed at Test Level III of MIL-R-22973. The position indicating assemblies shall be operated until 2000 hours have been completed or until there is a total of seven failures. The tests must be completed within four months of the date of the first contract delivery and repeated within thirty days. If a 500-hour MTBF has not been demonstrated, acceptance of position indicating assemblies will be withheld until corrective action is taken and its effectiveness is demonstrated. Position indicating assemblies used for these tests may be refurbished and delivered upon satisfactory completion of the tests.

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4.5 Test conditions -

4.5.1 Standard conditions - Unless otherwise specified, all tests required by this specification shall be made under the following conditions:

Temperature	Room ambient 25° ±5° C
Pressure	Normal atmospheric (approximately 29.92 inches Hg.)
Humidity	Room ambient up to 90 percent relative humidity
Input power	28.5 ±0.5 V DC

4.5.2 Test readings - Unless otherwise specified, before a test reading is taken, the indicator shall be vibrated using an electric vibrator set at 30 cps, 60 cps, or 120 cps with a maximum amplitude of 0.002 inch.

4.5.3 Attitude - Unless otherwise specified, the position indicating assembly shall be tested in its normal operating position.

4.6 Test methods -

4.6.1 Examination of product - Each position indicating assembly shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests.

4.6.2 Power off - With no power applied to the indicator, the wheel tabs shall rotate to the unlocked position and the flap tab shall rotate so that the legend "OFF" appears in the flap cut-out or an adjoining cut-out.

4.6.3 Insulation resistance - A potential of 200 volts (rms) alternating current at commercial frequency shall be applied between isolated pins and the case. The resistance between any two points thus chosen shall be not less than 20 megohms.

4.6.4 Operation -.

4.6.4.1 Wheels - Power shall be applied between pins H and I, F and I, and D and I. The respective tabs shall rotate until the legend "UP" appears. With power applied between pins G and I, E and I, and C and I, the respective tabs shall rotate so that the wheel image appears.

4.6.4.2 Flaps - A transmitter shall be connected to the flaps indicator and power shall be applied. The transmitter shaft shall be slowly rotated through an angle of 90 ±3 degrees. The flaps pointer shall move smoothly throughout its range of travel.

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4.6.5 Friction - The Operation test shall be repeated with the indicator tilted on an angle of approximately 45 degrees forward, backward, and to both sides. There shall be no sticking or friction which would prevent proper operation of the indicator.

4.6.6 Reduced voltage operation - The Operation test shall be repeated with the indicator connected to a 22.5 volt d.c. power source. The change in indication shall not result in exposure of more than 1/32 inch of the adjacent area when the indicator is viewed from the front of the dial.

4.6.7 Magnetic effect - This test shall be performed with all the wheels in the "Up-Locked" position. The indicator shall be properly connected and power applied. The indicator shall be revolved about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from the bar magnet. The compass shall have its compensating magnets removed and shall be set up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The indicator shall be revolved in a horizontal plane which is perpendicular to the axis of the bar magnet. The indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees. At each of these positions the indicator shall be rotated 360 degrees about its horizontal axis. The deflection of the compass at any of the specified positions shall not exceed three degrees. This test shall be repeated with no power applied.

4.6.8 High temperature operation - (This test may be combined with the High temperature exposure test at the discretion of the testing agency.) The position indicating assembly shall be properly connected except that no power shall be applied. The assembly shall then be subjected to an ambient temperature of 71° C (160° F) for a period of 4 hours. At the conclusion of the four hour period and while at 71° C, power shall be applied and the assembly shall be subjected to and shall meet the requirements of the Operation and Friction tests. Upon return to room temperature, the assembly shall again be subjected to and shall meet the requirements of the Operation and Friction tests.

4.6.9 Low temperature operation - (This test may be combined with the Low temperature exposure test at the discretion of the testing agency.) The position indicating assembly shall be properly connected except that no power shall be applied. The assembly shall then be subjected to an ambient temperature of -54° C (-65° F) for a period of 4 hours. At the conclusion of the four hour period and while at 71° C (para. 4.6.8) or -54° C (para. 4.6.9), power shall be applied and the assembly shall be subjected to and shall meet the requirements of the Operation and Friction tests. Upon return to room temperature, the assembly shall again be subjected to and shall meet the requirements of the Operation and Friction tests.

4.6.10 Position error - The transmitter shall be adjusted so that the flap indicator is down and the wheel indicator is in the "Up-Locked" position. The indicator shall then be held in the following positions:

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(a) Rotated from the normal operating position through 45 degrees to the rear (dial tilted 45 degrees backward from the vertical position).

(b) Rotated from the normal operating position through 45 degrees to the front (dial tilted 45 degrees forward from the vertical position).

(c) Rotated from the normal operating position through 45 degrees to the right (dial vertical).

(d) Rotated from the normal operating position through 45 degrees to the left (dial vertical).

The maximum change of the flap pointer from any one position to another shall not exceed five degrees and the wheel tabs shall not move sufficiently to expose the adjacent area. This test shall be repeated with the wheel indicator in the unlocked position. The wheel tabs shall not move sufficiently to expose more than 1/32 inch of the adjacent area.

4.6.11 Damping - The time required for the wheel tabs to come to rest after they have been released from the "Up-Locked" and "Down-Locked" positions shall not exceed three seconds.

4.6.12 Vibration (resonance) - The position indicating assembly shall be tested in accordance with Procedure XIII of MIL-E-5272. The pointer oscillation and the variation of the moveable tabs from their original position shall not exceed five degrees. At the conclusion of this test, the assembly shall be subjected to and shall meet the requirements of the Operation test.

4.6.13 Vibration (cycling) - The position indicating assembly shall be tested in accordance with Procedure XIII of MIL-E-5272. At the conclusion of this test, the assembly shall be subjected to and shall meet the requirements of the Operation test. In addition, no parts shall become loose nor shall damage to any part of the assembly result from this test.

4.6.14 Radio noise interference - The electrical system of the position indicating assembly shall be tested for conducted and radiated radio noise interference in accordance with MIL-I-6181.

4.6.15 Transmitter stop - The transmitter shall be connected to the indicator and test voltage shall be applied. A minimum torque of 25 inch-pounds shall then be applied to the transmitter shaft at the stop position. There shall be no open circuiting or other failure which would interfere with operation of the transmitter.

4.6.16 High temperature exposure - The position indicating assembly shall be tested in accordance with Procedure II of MIL-E-5272. At the conclusion of the temperature exposure and while still at the high temperature, power shall be applied and the assembly shall be subjected to and shall meet the requirements of the Operation and Friction tests.

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Upon return to room temperature, the assembly shall again be subjected to and shall meet the requirements of the Operation and Friction Tests.

4.6.17 Low temperature exposure - The position indicating assembly shall be tested in accordance with Procedure II of MIL-E-5272. While at -64° C the assembly shall be operated without exhibiting sticking or erratic characteristics. At the conclusion of the low temperature exposure and while at -54° C, power shall be applied and the assembly shall be subjected to and shall meet the requirements of the Operation and Friction tests. Upon return to room temperature, the assembly shall again be subjected to and shall meet the requirements of the Operation and Friction tests.

4.6.18 Transmitter cycling - The transmitter shall be mounted on a test panel and subjected to 200,000 cycles in such a manner that the moving contact of the transmitter travels over the resistance windings a distance of approximately 10 degrees on each side of the point at which either the up or down lock tap is taken (total movement approximately 20 degrees). The 200,000 cycles shall be complete reversals and shall be at the rate of 1,200 to 1,700 cycles per minute. There shall be no open circuiting or other failure which would interfere with operation of the transmitter.

4.6.19 Indicator cycling - Each indicator shall be subjected to 10,000 cycles of operation at a rate of 10 to 12 cycles per minute. A cycle of the flaps indicator shall consist of uniform movement from flaps "up" to flaps "down" and, back to flaps "up." A cycle of the wheel indicator shall consist of movement from the "unlocked" position, to the wheels "up and locked position," to the "unlocked" position, to the wheels "down and locked" position and back to the "unlocked" position. At the completion of the cycling, the indicator shall be subjected to and shall meet the requirements of the Operation test.

4.6.20 Salt spray - The assembly shall be subjected to salt spray in accordance with MIL-E-5272, Procedure I, for 50 hours. At the end of the 50 hour period, the assembly shall be examined. There shall be no corrosion or resultant damage from this test. The assembly shall then be subjected to and shall meet the requirements of the Insulation Resistance, Operation and Friction tests.

4.6.21 Mounting lugs - The case of the indicator shall be mounted face downward on the movable head of a suitable testing machine with the face of the indicator in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine. A load of 175 pounds shall be applied for one minute to each lug in a direction toward the front of the case. The lugs shall stand the applied load without fracture and there shall be no damage to any part of the indicator.

4.6.22 Internal examination - The case of the indicator and transmitter shall be opened and the internal mechanism shall be examined. Any deterioration or damage which could in any manner prevent the position indicating assembly from meeting functional operation or maintenance require-

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - For all levels, each indicator and transmitter shall be packaged in accordance with MIL-P-7936. Level A preservation shall be in accordance with MIL-P-116, Method IA, without preservative compound, using metal or fiberboard containers as specified by the procuring activity.

5.2 Packing - The indicators and transmitters shall be packed in accordance with MIL-P-7936, Level A, B, or C as specified in the contract or order (see 6.2).

5.3 Marking - The interior and exterior containers shall be marked as specified in MIL-STD-129.

5.3.1 Precautionary marking - The following precautionary markings shall appear on two opposite sides of each interior package and shipping container whenever practicable.

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended use - The position indicating assemblies covered by this specification are intended for use on aircraft to indicate the position of the wheels and flaps.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, Number and Date of this specification.

(b) The quantity and type of the position indicating assemblies desired.

(c) Levels of packaging (see 5.1) and packing (see 5.2) desired.

6.3 Data - For the information of Contractors and Contracting Officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower tier documents need not be prepared for the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractors Data Requirement List) which shall be attached to and made a part of the contract or order. For Navy contracts NavWeps form 4200/15 (Drawing Lists and Specifications Requirements) shall be attached where applicable.

6.3.1 Reliability - See 3.8.1.

6.3.2 Test reports - See 4.4.

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6.4 Drawing requirements - The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of MIL-D-70327.

6.5 Definitions -

6.5.1 Position indicating assembly - Whenever the term "position indicating assembly" is used, it refers to the indicator and transmitter, connected and ready for operation.

6.6 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for the opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity.

6.7 Precedence of documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.8 The outside margins of this specification have been marked "#" to indicate where changes (deletions, additions, etc.) from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in

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these notations. Builders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MO
Navy - WP
Air Force - MAAMA

Review activities:
Army - MO
Navy - WP
Air Force - MAAMA

User activity:
Navy - CG

Preparing activity:
Navy - WP
(Project No. 6610-0080)

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